



UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE



Academic Year 2021/2022 – First Year Examination – Semester I

SCS1205 – Computer Systems

TWO (2) HOURS

110

To be completed by the candidate

Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
3. Write your index number on each and every page of the Question paper.
4. Answer **ALL** questions.
5. There are **04** questions in **16** pages.
6. All 04 questions carry equal marks.
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
8. **Non-Programmable** calculators are **allowed**.

**For Examiner's use
only**

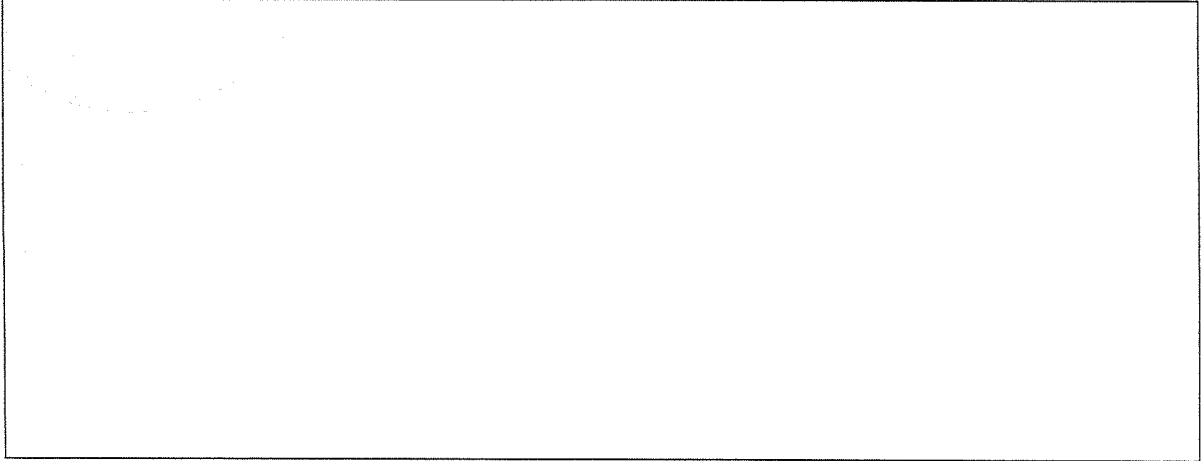
Question No	Marks
1	
2	
3	
4	
Total	

Index No:

Question 1

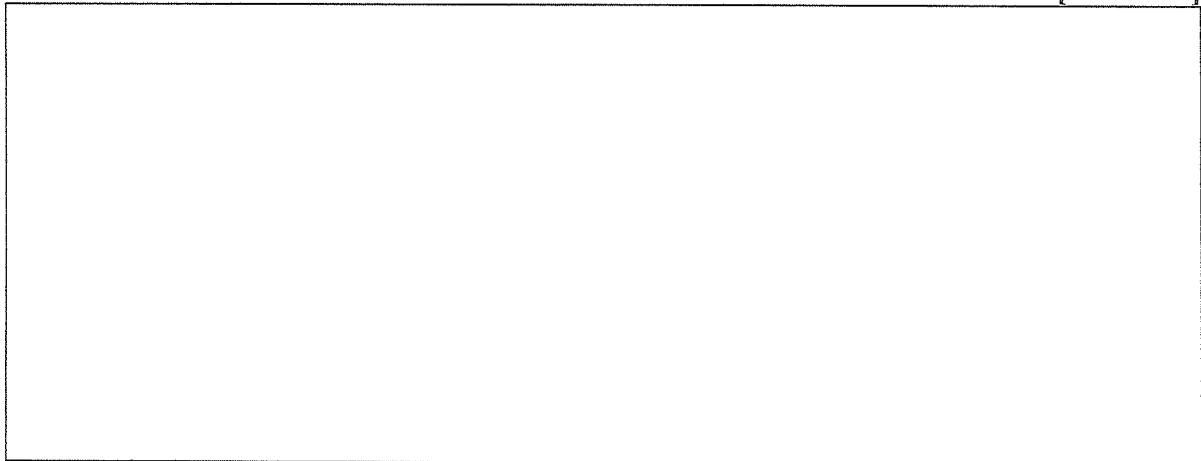
- A. In the past, there were multiple generations of computers produced. Write down three (03) techniques for classifying computers into generations.

[03 Marks]



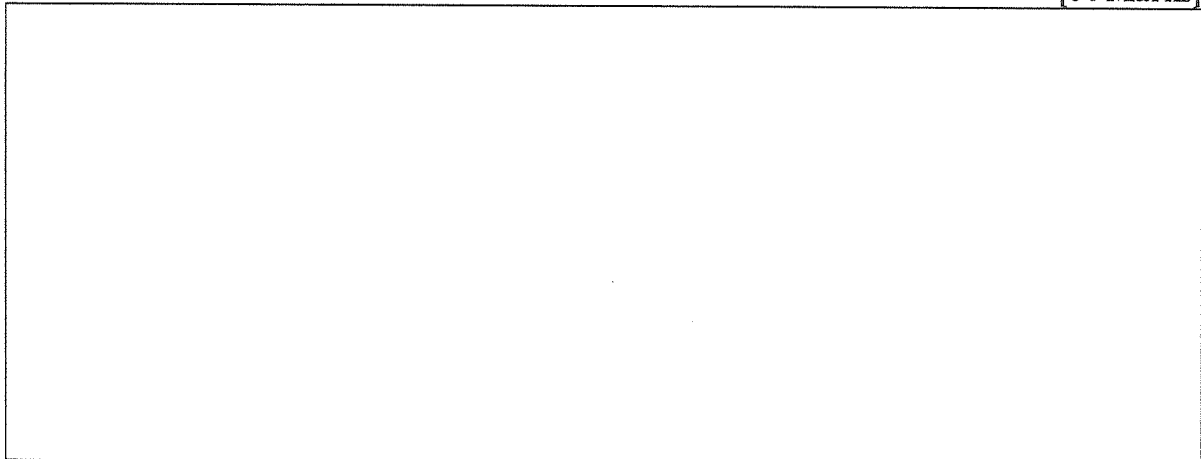
- B. What is the primary difference between a **second-generation** computer and a **third-generation** computer?

[03 Marks]



- C. What makes *Von Neumann Architecture* based computer systems unique?

[04 Marks]



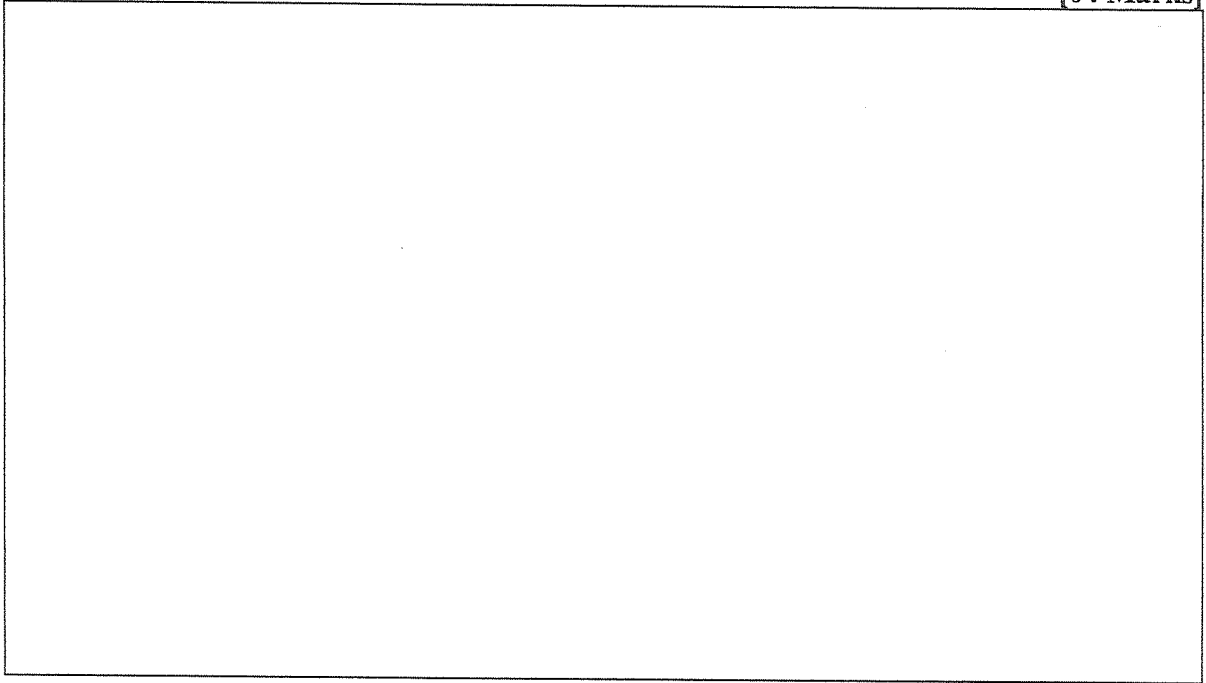
Index No:

D. Assume that a **sign bit**, a 7-bit **exponent**, and an 8-bit **mantissa** are all part of a 16-bit floating point representation.

i. What is the decimal equivalent of the 16-bit **floating point number** below?

0 1001001 11111111

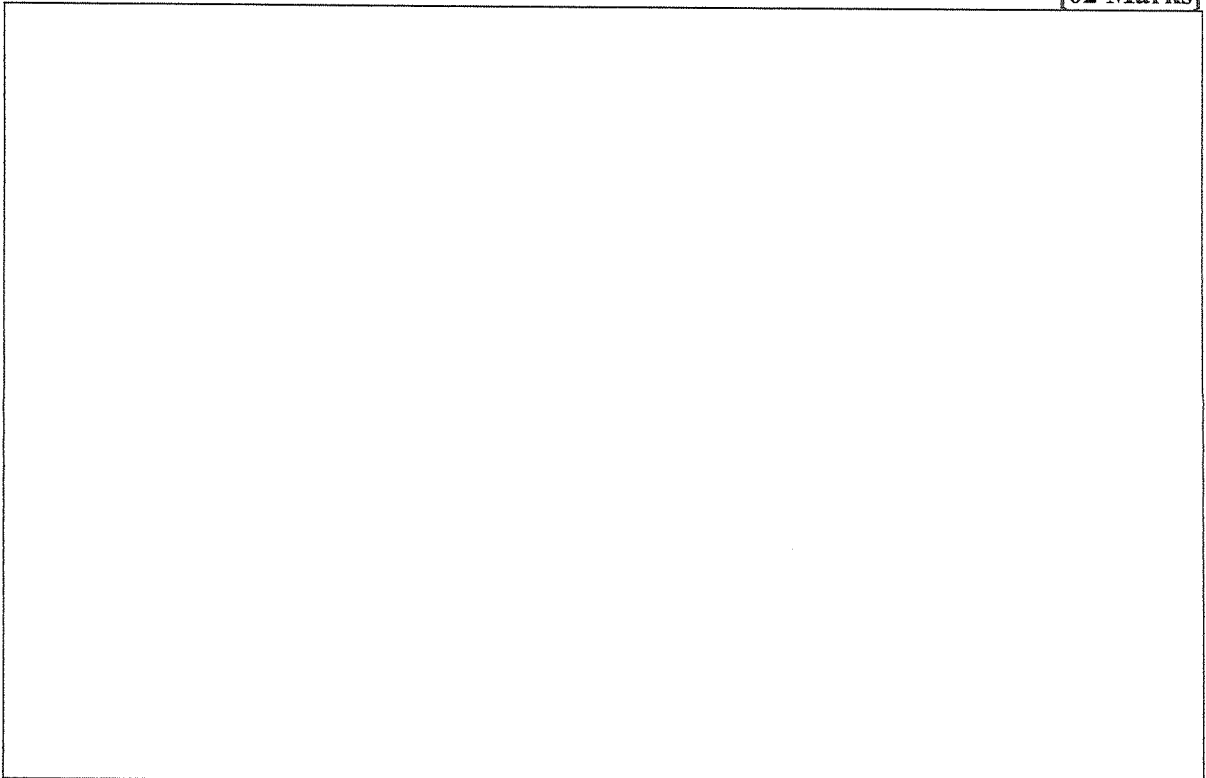
[04 Marks]



ii. What is the decimal equivalent of the 16-bit **floating point number** below?

0 1001010 00000000

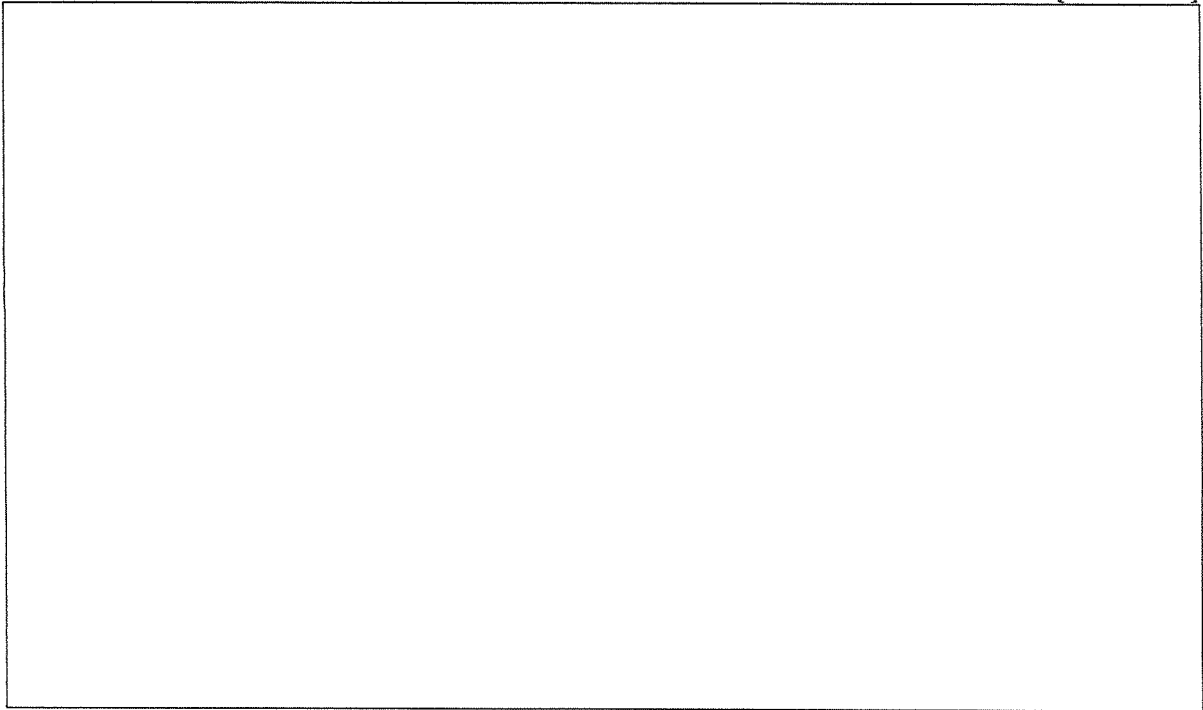
[02 Marks]



Index No:

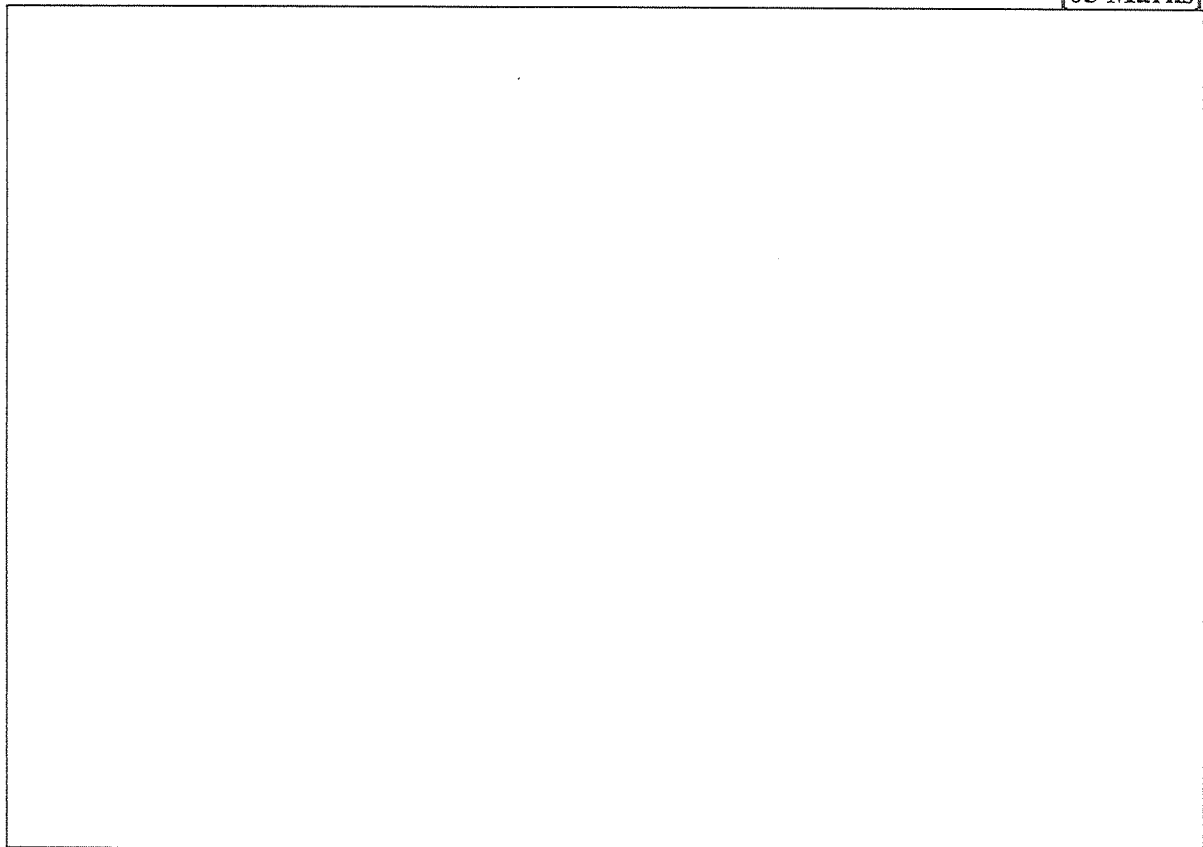
- iii. What is the 16-bit **floating point** representation of the **decimal** value +2045.5?

[04 Marks]



- iv. Have you noticed any contradictions between the solutions to Questions 1.D.i and 1.D.iii? Provide evidence to support your response.

[05 Marks]



Question 2

A. Map following two (02) Boolean functions F and G to the truth table given below.

- $F = (\bar{x}.\bar{y}.z) + (\bar{x}.y.z) + (x.\bar{y}.z) + (x.y.z)$
- $G = (x + y + \bar{z}).(x + \bar{y} + \bar{z}).(\bar{x} + y + \bar{z}).(\bar{x} + \bar{y} + \bar{z})$

[04 Marks]

x	y	z	F	G
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

B. Do you think two functions F and G in the Question 2.A above are logically equivalent? State your answer with reasons.

[02 Marks]

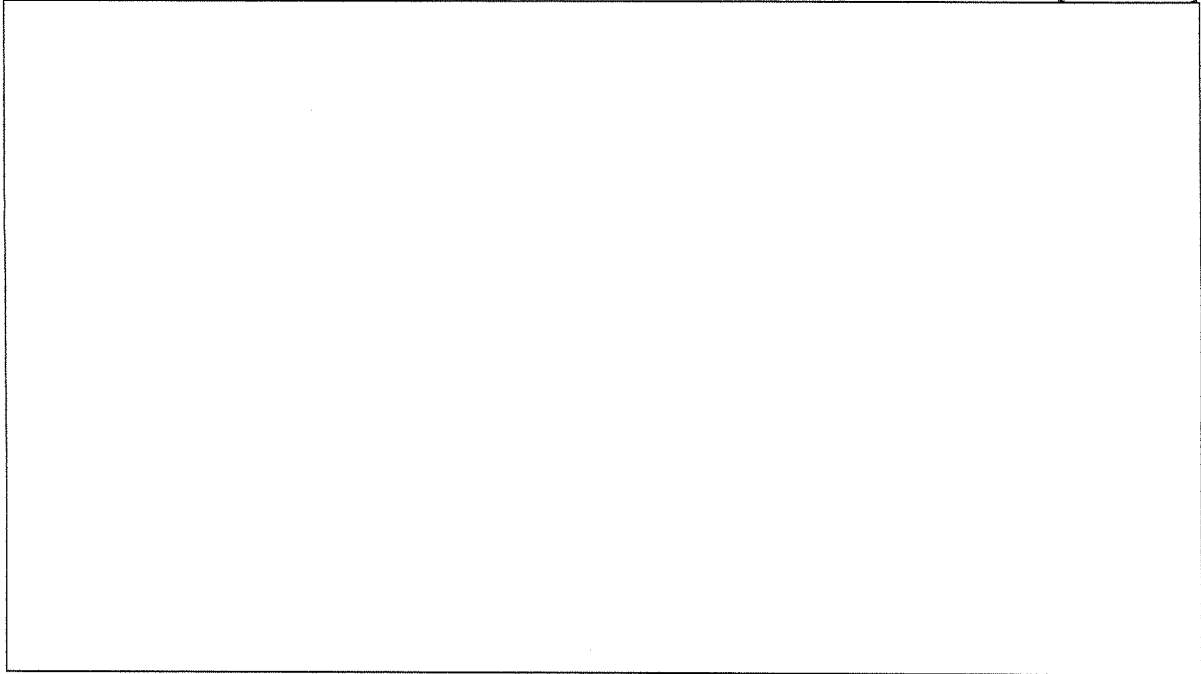
C. Derive the logical complement of the Boolean function G (i.e. $\bar{G}$) directly from the truth table above in Question 2.A, in the form of **Product of Maxterms**. Simplify the expression for $\bar{G}$ as much as possible **only using Boolean algebraic laws** by showing your simplification steps clearly.

[04 Marks]

Index No:

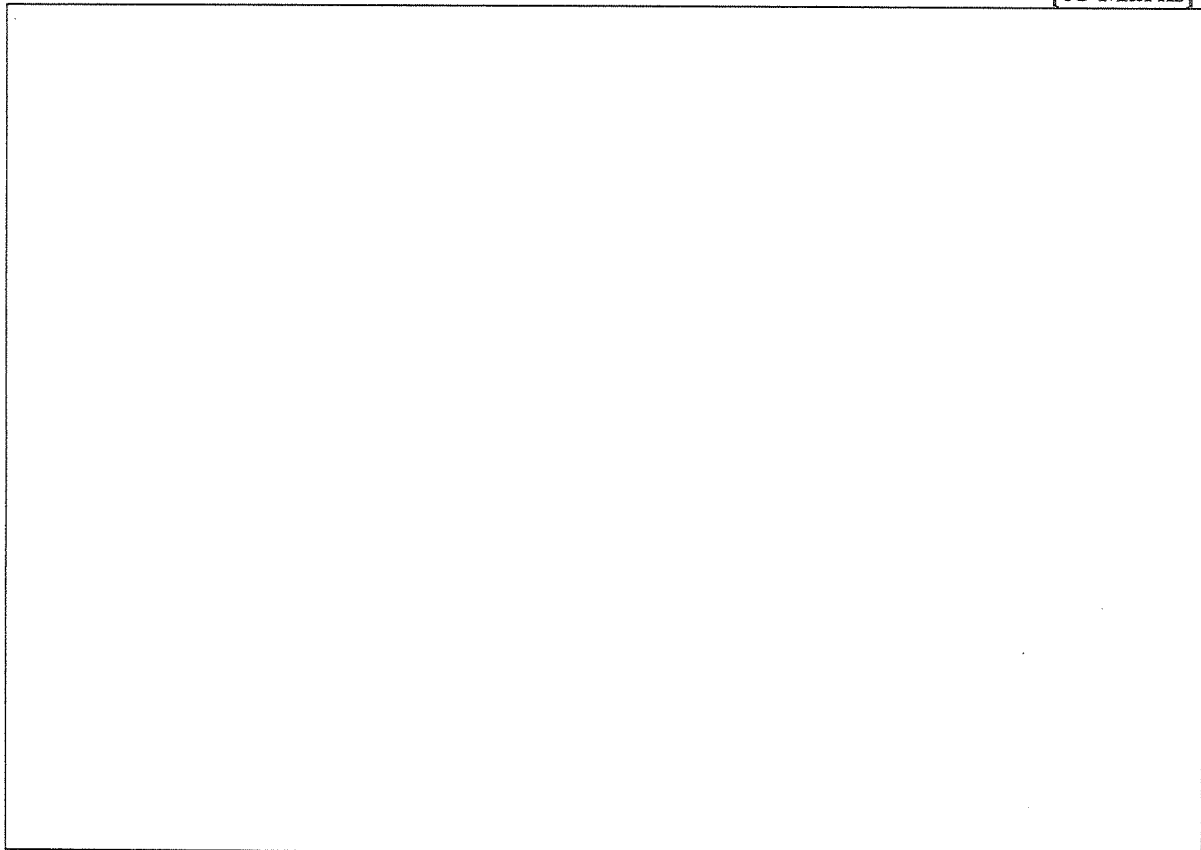
- D. Draw a **diagram** and show how you can implement a circuit to perform the Boolean function F in the above Question 2.A **only using a 3×8 Decoder** and an extra logic gate. **Clearly label the Decoder's output channels.**

[04 Marks]

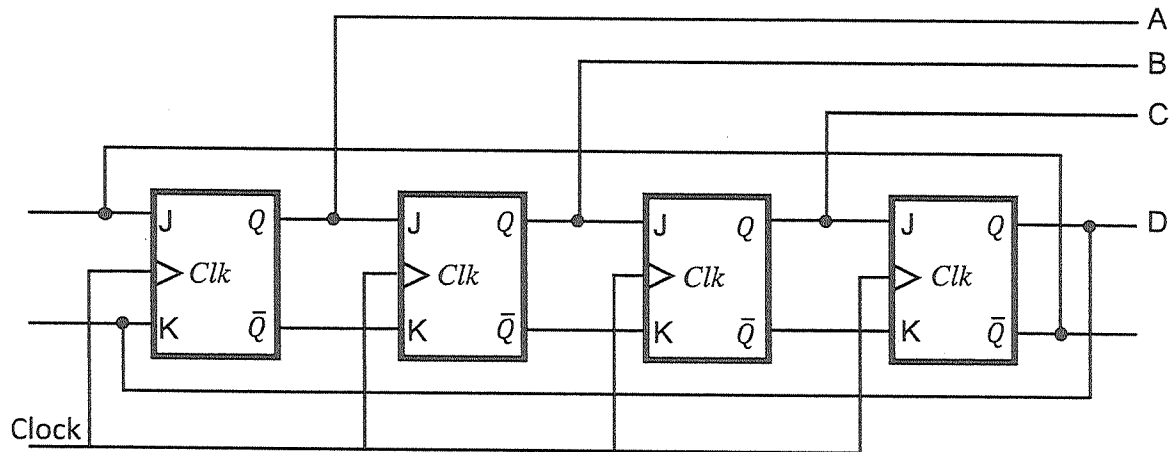


- E. Draw a Full-Adder circuit **only using NAND gates.**

[05 Marks]



- F. Consider a hypothetical sequential logic circuit which consist of four **positive edged J-K flip flops** as given below.

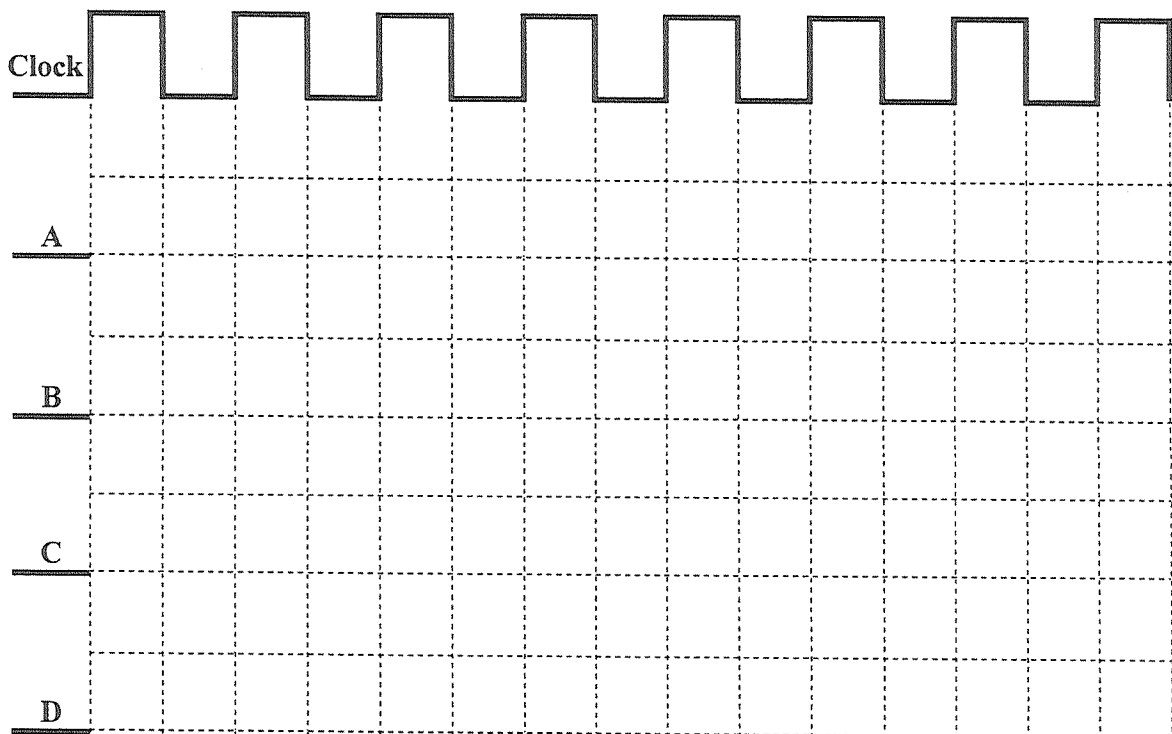


Assume that all A, B, C, and D are in logical state 0 at the first clock pulse.

Draw the **timing diagram** for A, B, C, and D outputs on the *dashed line drawing* given below based on the available clock pulse pattern.

Note that the initial states of all A, B, C, and D are drawn as state 0 at the first clock pulse.

[06 Marks]

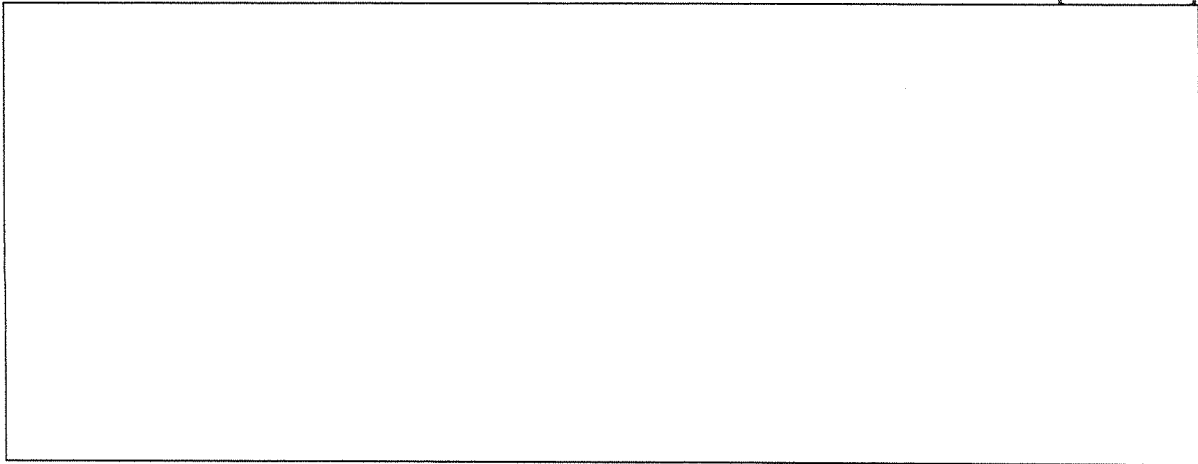


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Question 3

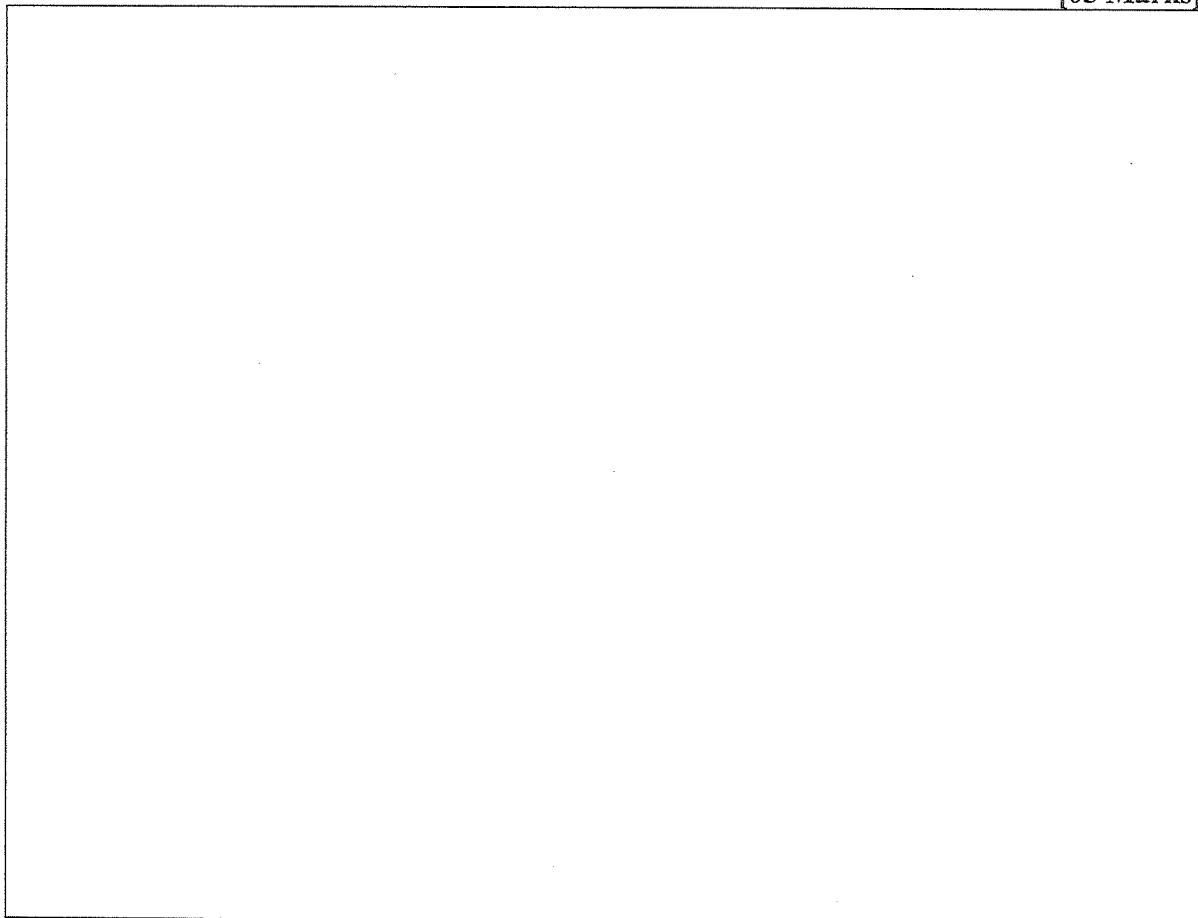
A. What are the two (02) main components of a CPU?

[02 Marks]



B. We have two machines A and B with different implementations of the same Instruction Set Architecture (ISA). Machine A has a clock cycle time of **100 ns** and a Cycles Per Instruction (CPI) of 1.0 for program **P**; Machine B has a clock cycle time of **50 ns** and a CPI of 1.8 for the same program. Assume that the number of instructions in the program **P** is 1,000,000. Which machine executes program **P** faster? **Clearly show** your reasoning with calculations and steps.

[05 Marks]



C. Consider the below *pseudocode* segment.

```
100 sequential Instructions;  
For (i=0; i<50; i++) do  
    18 sequential Instructions;  
EndFor
```

i. How many **static** instructions will be executed within this code segment?

[01 Mark]

ii. How many **dynamic** instructions will be executed within this code segment?

[01 Mark]

D. Briefly explain two (02) critical performance techniques employed by the RISC architecture.

[02 Marks]

E. *Stack architecture* is one of the instructions set architectures.

i. State two (02) advantages of the stack architecture.

[02 Marks]

Index No:

ii.

a. Convert the below expression to *postfix notation*.

$$\{(x - y)/(x + y)\}/(x * y)$$

b. Write down the execution steps for a stack-based instruction set architecture for the above expression.

[04 Marks]

- F. Suppose there exist a machine **M** where instructions are **16-bit** long and word size of the machine is **8 bits** and byte addressable. Instructions are in the form of:

<opcode> <register> <memory address>

Here, **first 4 bits** are allocated for the **opcode**, next **4 bits** are for the **register** and the **remaining bits** are for the **operand's memory address**.

Following are some listed opcodes for different types of instructions in **M**.

L R, A	<i>Load the register R with the content of memory cell with the address A</i>
LI R, I	<i>Load the register R with the value I in hexadecimal</i>
LR R1, R2	<i>Load the register R1 with the content of the register R2</i>
ST R, A	<i>Store the content of register R to the memory cell whose address is A</i>
ADD R0, R1, R2	<i>Add the two numbers in registers R1 and R2 and place the result in R0</i>
OR R0, R1, R2	<i>Perform the logical OR with the bit patterns in R1 and R2 and place the result in R0</i>
XOR R0, R1, R2	<i>Perform logical XOR with the bit patterns in R1 and R2 and place the result in R0</i>
AND R0, R1, R2	<i>Perform logical AND the bit patterns in R1 and R2 and place the result in R0</i>
JMP R, A	<i>Jump to the instruction located in the memory cell with the address A if the bit pattern in R is equal to R0</i>
HALT	<i>Halt the execution</i>

Write down the machine code instruction sequence to execute the following program segment **only** using the above instructions.

Assume that **Total**, **Count**, and **Index** are variables that refer the memory addresses **0x70**, **0x80**, and **0x90** respectively and the **Program Counter (PC)** is initialized to **0x16**.

```
Total=0;
Count=10;
for (Index=0; Index<Count; Index++) {
    Total = Total - Index;
}
Return;
```

[08 Marks]

Index No:

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Question 4

- A. In a hypothetical computer system with an 8 Bytes of **direct mapped** cache and a 1024 Bytes of main memory, the CPU refers twelve (12) addresses in a **sequence** as given in the following table. The cache line size is two (02) Bytes, and the memory is byte addressable.

Note: Assume that all the addresses are in binary representation and the cache was empty before referring all these addresses.

Sequence No.	Address	Associated Data
1	0000 0000 1010	A
2	0000 0000 1011	B
3	0000 0010 1100	C
4	0000 0001 0010	D
5	0000 0000 1000	E
6	0000 0010 1111	F
7	0000 0000 1001	G
8	0000 0001 0001	H
9	0000 0010 1101	I
10	0000 0001 0000	J
11	0000 0001 0011	K
12	0000 0010 1110	L

How many **bits** should be reserved as the **offset component** of an address in this cache configuration? **Clearly show** your calculations and steps with reasoning.

[03 Marks]

- B. How many bits should be reserved as the **index component** of an address in the cache configuration described in above Question 4.A? **Clearly show** your calculations and steps with reasoning.

[03 Marks]



- C. How many **bits** should be reserved as the **tag** component of an address in the cache configuration described in above Question 4.A? **Clearly show** your calculations and steps.

[02 Marks]

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- D. By filling the **cache layout diagram** given below with appropriate Tag, Index, Offset, Data, and Valid Bit values, show what would be the **final cache arrangement after referring all addresses** in the table in the Question 4.A.
Keep a cell empty in the diagram if there is no data mapped to it.

[08 Marks]

Tag	Index	Offset	Data	Valid Bit

- E. To transform the above cache layout diagram in Question 4.D to a **2-Way Set Associative cache**, what should be the **size** of a cache set? Give reasons for your answer.

[03 Marks]

- F. Assume a **4-Way Set Associative** cache that uses *First-In-First-Out* (FIFO) cache replacement policy. The CPU consecutively refers to four (04) memory addresses A, B, C, and D in that order which results in a cache miss at each address reference. Is there a possibility that the cache reference to address D above could **replace** the contents loaded to the cache by one of the references A, B, or C? Give reasons for your answer.

[03 Marks]

Index No:

- G. The CPU refers the address E immediately after the four cache references in the cache system described in **Question 4.F**, which results in a **cache miss** (i.e., new cache entry for E is created). Assume addresses A, B, C, D, and E are mapped to the same cache set. In this circumstance, what is the **address** from A, B, C, D, and E that should be referred next to result a **cache miss** again? Give reasons for your answer.

[03 Marks]
